

Project Drishti: Comprehensive Progress Report, HCI Evaluation, and Future Architecture Roadmap

Date: July 29, 2026

Project Scope: Low-Cost Predictive Gaze-Typing Web Interface for Bengali Language Accessibility

Core Fields: Computer Vision, Human-Computer Interaction (HCI), Computational Linguistics

1. Project Aims and Executive Objectives

Project Drishti is engineered to address a critical gap in assistive technologies for individuals suffering from severe motor-neuron impairments, quadriplegia, or non-verbal conditions (such as ALS or advanced spinal cord injuries). The primary goal is to provide an completely non-invasive, high-throughput, and economically accessible Bengali text-composition system. By leveraging standard commodity consumer webcams, the system eliminates the extreme cost barriers associated with proprietary active infrared eye-tracking hardware.

- **Hardware Dematerialization:** Transition eye-gaze tracking from expensive specialized sensors (\$200+) into a zero-cost software execution framework utilizing basic laptop webcams.
- **Multi-tier Interface Accessibility:** Seamlessly integrate real-time Avro Phonetic Transliteration alongside high-dimensional dense vector models to optimize text entry workflows.
- **HCI Ergonomic Optimization:** Minimize physical oculomotor strain and visual searching overhead through the rigorous application of established interaction principles (e.g., Fitts's Law).

2. Technical Progress and Architecture Overview

The current software ecosystem is split into a high-performance Python computer vision server and a highly reactive React.js frontend interface communicating over a low-latency local WebSocket pipeline.

System Layer	Core Technologies	Current Implementation Status
Perception / CV	Google MediaPipe, OpenCV, Python AsyncIO	Extracts 468-point 3D facial mesh, isolates iris boundaries (468-473), and pipes normalized coordinates stream via WebSockets at 30-60 FPS.
Signal Processing	Bivariate Kalman Filter, Polynomial Matrix	Smooths out frame-by-frame coordinate pixel jitter and applies a customized 9-point regression calibration model to handle screen warping.
Interaction UI	React.js, Tailwind CSS, Avro Phonetic Engine	Displays a static, high-contrast QWERTY canvas layout with expanding progress triggers to bridge the Gulf of Evaluation. Converts Latin layouts into native Bengali script.
Semantic NLP	LaBSE (BERT Encoder), Qdrant Vector DB	Runs localized dense vector embeddings query maps in memory, rendering full-sentence autocomplete recommendations into a reactive sidebar layout.

3. Key Empirical and Research Findings

Through rigorous review of foundational computer vision and HCI literature (including top-tier benchmarks like MPIIGaze and state-of-the-art 3DMM spatial architectures), the following insights have been integrated into our project parameters:

1. **The Spatial Muscle Memory Advantage:** Validated that keeping a static, standard layout configuration outperforms dynamic layout re-arrangements (such as Flex-Tree). Preserving key positions allows users to build spatial muscle memory, cutting searching delays drastically.

2. **The Boundary Placement Paradigm:** Confirmed that positioning high-frequency modification triggers (like Space, Delete, or the AI Prediction Sidebar) at the extreme outer boundaries of the display screen provides a physical buffer that accommodates calibration drift seamlessly.
3. **Phonetic Distance Compression:** Proved that combining a standard layout with real-time phonetic transliteration maps natively cuts down the required ocular trajectory path. This significantly lowers the overall Index of Difficulty (ID) defined by Fitts's Law when composing non-Latin scripts.

4. Critical Bottlenecks and System Drawbacks

Despite structural successes, testing under natural conditions has highlighted three major computational and architectural constraints:

- **Environmental Illumination Sensitivity:** Raw appearance-based eye patches remain highly vulnerable to extreme lighting shifts, asymmetric room shadows, and fluctuating display glares, which can degrade coordinate accuracy.
- **Oculomotor Dwell Fatigue:** Forcing a fixed temporal look-dwell filter (e.g., 1200ms) introduces a strict trade-off: short windows cause rampant accidental selections (the Midas Touch problem), while longer windows cause muscle strain.
- **Sub-Word Tokenization Fragment Breaks:** The pre-trained LaBSE encoder drops vector lookup accuracy when confronted with incomplete text strings or minor spelling variations, sometimes returning irrelevant completions.

5. Future Architecture Roadmap and Upcoming Features

To establish deep academic novelty and transition Project Drishti into a clinically robust solution, the next engineering phase introduces advanced features across three distinct engineering tiers:

A. Near-Infrared (IR) Vision Hardware Integration

The system will bypass standard RGB illumination problems entirely by moving to a high-speed (120 FPS) monochrome Near-IR USB web camera module stripped of its IR-cut filter and wrapped in an invisible 850nm LED bar. This allows the backend to capture ultra-stable pupil geometry independent of room lighting conditions. The backend will implement a **Pupil-Glow Vector Matrix**, measuring the distance between the pupil center and the stationary LED glint reflection on the cornea to neutralize head-pose drift entirely.

B. Biomedical Adaptations and Cognitive Gating

- **Asymmetric Muscle Gain Tuning Filter:** Evaluates tracking variance between both eye meshes independently. If a user exhibits asymmetrical motor drop or palsy, the script shifts reliance to the dominant eye and alters Kalman variables dynamically.
- **Cognitive Dilation Dwell Controller:** Tracks relative changes in the pupil area to capture the involuntary dilation reflex that occurs when a human intends to select a target. This allows the system to drop selection delays from 1200ms to 600ms dynamically based on user intent.
- **Physiological Ocular Blink Masking:** Integrates an internal state buffer that freezes the cursor coordinates during natural blinks (<250ms), preventing data drops from interrupting look-dwell selections.

C. IoT Environmental Control Hub and Secure Ledger Log

The coordinate tracking distribution loops will expand to drive a **Smart-Grid Environmental Control Suite**. By looking at a menu button, the user can toggle the interface to control smart home components (such as lights or fans) locally and securely. Furthermore, critical distress events triggered via the Emergency SOS grid will be hashed instantly using SHA-256 and stored in an immutable localized event database. This provides an unalterable history that caregivers or clinicians can review to verify patient distress calls.